

Brian Nielsen

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EDUCATION

University of Florida - GPA: 3.92

Bachelor of Science in Computer Science, Minor in Electrical Engineering

Gainesville, FL

Aug. 2021 – May 2025

EXPERIENCE

Software Engineer

Aug. 2024 – May. 2025

Design Project Sponsored by Florida Power and Light

Gainesville, FL

- Built a Raspberry Pi hazard detector using **multiprocessing** to isolate 100ms ML inferences from UI actions.
- Wrote **bash scripts** for boot setup/tuning and developed UI components to display CPU and temp metrics.
- Sharpened **communication skills** through team/stakeholder meetings, presentations, and documentation.

Undergraduate Research Assistant

May 2024 – Aug. 2024

SeCurity and Quality AssuraNce (SCAN) Lab, University of Florida

Gainesville, FL

- Researched **AI-driven** PCB identification and assurance through ML models such as **YOLOv8** and **CoCa**.
- Studied the **IC supply chain** and production, diving into die design, packaging, and fab vs. fabless facilities.
- Explored **semantic segmentation** to combine Microscope, X-Ray and Sonic images into one classification model.

Circuits 1 Teaching Assistant

Jan. 2024 – May. 2025

University of Florida

Gainesville, FL

- Taught the basics of **op-amps**, **filters**, and **chip manufacturing** to lab sections with a total of **50+** students.
- Graded lab reports, hosted office hours, and taught a 2 hour in person lab every week.

Technical Lead

Aug. 2022 – Dec. 2023

Open Source Club, University of Florida

Gainesville, FL

- Managed a GitHub repository which involved creating issues, merging pull requests, and completing code reviews.
- Wrote professional documentation so a club of **1,000+ members** can understand our code-base.
- Refined teamwork skills through bi-weekly meetings with the project's team.

PROJECTS

R-NES | Rust

May. 2025 – Present

- Developed a NES emulator in Rust, covering **151** CPU instructions, **interrupt handling**, and 110+ test cases.
- Built proficiency in Rust **lifetimes**, **closures**, and **borrow checking** through **low-level systems modeling**.
- Reverse-engineered NES **memory-mapped I/O**, including PPU register mirroring and cartridge mappers.

Study Sesh — (Swamphacks X) | React, JavaScript, Django, MongoDB, Auth0, Python, Git

Jan. 2025 - Feb. 2025

- Established a full stack web-app in **22 hours**, providing UF students a tool to locate study partners.
- Made with React, Django RESTful API, and a MongoDB database of **6,000+** scraped UF course entries.
- Implemented **Auth0** for JWT authentication, using a **three-way token** system to enhance endpoint protection.

Digital Reverb Plugin | C++, JUCE, Git

Nov. 2024 – Mar. 2025

- Engineered a reverb plugin in **JUCE** with diffusion, delays, and a **Hadamard matrix** for a customizable sound.
- Created adjustable reverb parameters to customize delay timings and diffusion attributes.
- Overcame C++ audio programming challenges, delivering a functional reverb plugin optimized for guitar.

UFmyMusic | C++, TCP

Aug. 2024 – Dec. 2024

- Produced a **TCP** application layer protocol for syncing music files between server/clients.
- Utilized the **OpenSSL** library to hash files, ensuring **100%** accurate duplicate song file detection.
- Leveraged **multi-threading** and **semaphores** to allow multiple clients to interact with the server simultaneously.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, Rust, SQL, Typescript, Javascript, MATLAB, Golang, JSON, Linux Scripts, HTML

Frameworks: React-Native, Node.js, Django, NextJS, NodeJS, Auth0, MongoDB

Developer Tools: Git, Virtual Box, AWS, AWS Lambdas, Cypress, JUnit, VS Code, Visual Studio, Eclipse

Protocols/Libraries: I2C, SPI, UART, gin, pandas, NumPy, Matplotlib, TensorFlow

Classwork: Data Structures & Algorithms, Networking Fundamentals, Operating Systems, Intro to SWE, Signals and Systems, Foundations of DSP, Electronic Circuits, Physics of EE